DANIEL DAVIS

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Research focus: Machine learning and Bayesian inference for ill-posed inverse problems on large, noisy sensor data at high performance computing (HPC) scale. PhD candidate with 8+ years of industry R&D at PETRONAS before the PhD.

Full name: Mohammed Daniel Davis bin Muhammed aka Daniel Davis.

EDUCATION

Colorado School of Mines, Golden, CO, United States — Aug 2023 – Expected 2028
PhD in Geophysics, Minor in Data Science and Machine Learning. GPA 3.9/4.0
Center for Wave Phenomena (Advisor: Prof. Jeffrey Shragge)
Dissertation: *Wavefield Gradiometry for Ultra-Low-Frequency Interferometric Ocean Bottom Node (OBN) Data*

Imperial College London, United Kingdom — Sep 2013 – Sep 2014
MSc in Petroleum Geophysics. Malaysian Government Postgraduate Scholarship
Thesis: *Rank-Reduction Iterative Cadzow Method for Deblending* (Advisor: Dr. Thomas Elboth, Viridien)

Universiti Putra Malaysia, Malaysia — May 2008 – May 2012
BSc in Physics

SELECTED PUBLICATIONS

1. **Davis D.**, Shragge, J., de Ridder, S., Girard, A.J., and Pandey, A. (2026). Interferometric frequency-domain gradiometry. *Geophysical Journal International*, 245(3), ggag146. doi:10.1093/gji/ggag146
2. **Davis, D.**, Shragge, J., and Garvey, S. (2026). Trans-dimensional Bayesian inversion of interferometric frequency-domain gradiometry. *International Meeting for Applied Geoscience & Energy 2026 (IMAGE 2026) (Society of Exploration Geophysicists (SEG) / American Association of Petroleum Geologists (AAPG))*. Poster presentation.
3. **Davis, D.**, Shragge, J., de Ridder, S., and Girard, A.J., (2025). Ambient seismic frequency-domain gradiometry. *International Meeting for Applied Geoscience & Energy 2025 (IMAGE 2025) (SEG/AAPG)*. Poster presentation.
4. Rusmanugroho, H., Li, J., **Davis Muhammed, M.D.**, and Sun, J. (2022). 3D velocity model building based upon hybrid neural network. *International Meeting for Applied Geoscience & Energy 2022 (IMAGE 2022) (SEG/AAPG)*, 1634–1638. doi.org/10.1190/image2022-3745433.1
5. **Muhammed, M.D.D.**, Isa, M.N., and Mishra, S. (2022). Comparison between the pseudo-analytical and finite difference method for seismic modelling and imaging. *Offshore Technology Conference Asia 2022 (OTC 2022)*. Poster presentation.
6. **Muhammed, M.D.D.**, Zahir, M.H.M., Ghazali, A.R., and Rahim, M.F.A. (2020). Imaging of fiber optics DAS VSP installed behind casing and with an inflatable liner. *EAGE Workshop on Fiber Optic Sensing, Asia Pacific*. doi:10.3997/2214-4609.202070028
7. **Muhammed, M.D.D.**, Rahim, M.F.A., and Ghazali, A.R. (2019). Integrated shallow gas imaging using fiber optics DAS VSP and OBN. *81st European Association of Geoscientists and Engineers (EAGE) Conference & Exhibition*. doi:10.3997/2214-4609.201901251

Publishes as Davis, D., Davis Muhammed, M.D., and Muhammed, M.D.D. Full list on Google Scholar.

RESEARCH EXPERIENCE

Colorado School of Mines, PhD researcher, Colorado, United States — Aug 2023 – Present

- **Interferometric frequency-domain gradiometry (I-FDG):** Derived and implemented a method that extracts coherent sub-2 Hz wavefields from long-duration ambient ocean-bottom recordings by cross-correlation, then estimates local wave velocities in the frequency domain. Stacking virtual-source gathers raises signal to noise ratio (SNR) and depth sensitivity over conventional time-domain gradiometry. Validated on a 3-D synthetic salt model with strong agreement with dispersion-curve analysis (First author, Geophysical Journal International, 2026).

- **Trans-dimensional Bayesian inversion:** Building a reversible-jump Markov Chain Monte Carlo (MCMC) sampler in Python where the number of model layers is itself inferred, mapping gradiometry measurements to depth profiles of shear velocity with full posterior uncertainty. Preliminary results agree closely with synthetic salt models.
- **HPC pipeline:** Built a 3-D Bayesian surface-wave inversion pipeline on the Waveform and Wendian cluster with SLURM array jobs, 3-D radial basis function (RBF) interpolation and interactive visualizations of the posterior volumes.

INDUSTRY EXPERIENCE

PETRONAS, Research geophysicist, Kuala Lumpur, Malaysia Feb 2015 – Aug 2023

- **Machine learning (ML) for subsurface property estimation:** Led a project predicting 3-D velocity fields from seismic data with a convolutional neural network (CNN) constrained by a recurrent neural network (RNN), combined with decision trees. On SEG's synthetic SEAM and Gulf of Mexico Block 26 field data it reached up to 97% accuracy with 80% less training data (IMAGE 2022 Conference).
- **Simulation-driven survey design:** Led wave-equation modeling (pseudo-analytical vs finite-difference) for an ultra-deepwater salt play, Gabon Block F12. The recommended acquisition geometry saved an estimated USD 200k per day (OTC Asia 2022 Conference).
- **Large-array sensor processing:** Managed a USD 1M+ fiber-optic distributed acoustic sensing (DAS) project at Bokor field, denoised 50,000+ source shots contaminated by production noise (f-x filtering) and imaged them with reverse time migration (RTM) with multiples, integrated with OBN data at zero production downtime (EAGE 2019, 2020 Conference).
- **Sensor development:** Co-developed a multi-parameter fiber-optic sensing system (acoustic, temperature, pressure, strain) with Neubrex Co. Japan, field-trialed at the University of Calgary Containment and Monitoring Innovation (CAMI) site, projected savings of USD 500k per installation. Won PETRONAS's Project Innovation & Novelty Award 2019.
- **Signal processing and ML for oil and gas production:** Spectral-decomposition and image-processing workflows identified 230 million barrels of projected leads (Namibe Basin, Angola), deployed an ML multi-attribute pattern-recognition tool used by asset teams for prospect de-risking.
- **Software:** Worked with software engineers on Schlumberger Petrel plugin development and debugging, administered the department's Microsoft Team Foundation Server (TFS) for version-controlled code.

Viridien (formerly CGG), Geophysicist (secondment), Kuala Lumpur, Malaysia Jun 2018 – Dec 2018

- Processed DAS vertical seismic profiling data from three producing wells using Viridien's Geovation software.

Viridien (formerly CGG), R&D student researcher, Oslo, Norway May 2014 – Aug 2014

- Designed simultaneous-source deblending in MATLAB with sliding-window time-domain singular value decomposition (SVD) and frequency-domain Cadzow filter.

CERN, Summer student, Geneva, Switzerland May 2012 – Aug 2012

- Ran Monte Carlo simulations of particle kinematics and background processes for the newly discovered Higgs boson and estimated its expected mass from simulated events.

SKILLS

ML & statistics: Bayesian inference, MCMC, reversible-jump (trans-dimensional) sampling, uncertainty quantification, Monte Carlo simulation, CNNs, RNNs, decision trees.
Inverse problems & simulation: Full waveform inversion (FWI), reverse time migration (RTM), finite-difference and pseudo-analytical wave modeling, surface-wave gradiometry, seismic interferometry.
Signal processing: Singular value decomposition (SVD), Cadzow filter, *f-x* filtering, *f-k* filtering, spectral decomposition, denoising large sensor arrays
Engineering & compute: Python (NumPy, SciPy, Plotly), MATLAB, Madagascar/SCons, SLURM and HPC clusters, Git, Claude Code (daily use for research software)

AWARDS

PETRONAS Project Innovation & Novelty Award (2019), PETRONAS Focused Recognition Award (2019, 2020, 2021), Malaysian Government Postgraduate Scholarship (2013)